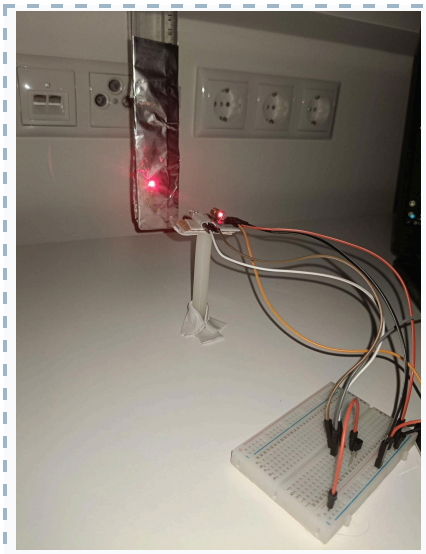
NON-CONTACT OPTICAL VIBRATION SENSING

Frequency Analysis Using Reflected Laser Light, Embedded Data Acquisition and FFT

Experimental Project Report



Hossam
GIU Berlin
October 2026

Project focus: Photonic sensing • Embedded acquisition • Signal processing • Experimental characterization

Abstract

This project investigates a low-cost, non-contact optical method for measuring mechanical vibration frequencies from reflected laser light. A continuous laser beam was directed onto a vibrating target and the reflected beam was received by a light-sensitive detector connected to an embedded acquisition system. Small mechanical motion produced measurable variations in received optical intensity. The signal was sampled at a nominal 1 kHz rate, buffered locally to avoid serial-timing disturbances, transferred to Python, and transformed from the time domain into the frequency domain using the Fast Fourier Transform (FFT). With 500 analyzed samples, the spectral bin spacing was 2 Hz and the theoretical Nyquist limit was 500 Hz.

The prototype was evaluated using four experimental cases: a cantilevered ruler, the same ruler with an added €2 coin, a DC motor, and a piezo/buzzer excitation source. The ruler experiment demonstrated structural resonance; the added-mass test explored the expected reduction in resonance frequency; and the motor experiment produced a dominant component around 52 Hz together with higher-order components. The development process also exposed real measurement challenges, including a weak detector dynamic range, alignment sensitivity, ADC headroom limitations, startup transients, and a strong 50 Hz mains-related component. Grounded conductive shielding reduced the observed 50 Hz interference by approximately 70% in the tested configuration.

The main outcome of the work is not only the measured frequency peaks, but the complete experimental chain: optical transduction, analog acquisition, embedded timing, interference investigation, signal processing, and interpretation of mechanical behavior. The system is intentionally presented as an experimental demonstrator rather than a calibrated laser vibrometer. Future improvements are proposed around a dedicated transimpedance amplifier, differential optical detection, hardware-timer sampling, anti-aliasing, and calibrated displacement measurement.

Key technical parameters Nominal sampling rate: 1 kHz | Stored samples: 550 | Analyzed samples: 500 | FFT bin spacing: 2 Hz | Theoretical Nyquist frequency: 500 Hz | ADC: 10-bit range (0–1023 in the used configuration).

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1. Motivation and Objectives

Mechanical vibration carries information about the dynamic behavior of structures and machines. Resonant frequencies can change when mass, stiffness, boundary conditions, or operating conditions change. Conventional measurements often use contact sensors such as accelerometers. Optical methods provide a different approach: motion can be inferred from changes in light without attaching an electrical sensor directly to the measured object.

The objective of this project was to investigate how far a simple optical system could be pushed using inexpensive components and software-based analysis. The emphasis was on experimental understanding rather than reproducing professional laser vibrometry. Specifically, the project aimed to:

- Convert small mechanical vibration into a measurable optical intensity variation.
- Acquire the detector signal at a controlled and repeatable sample rate.
- Extract dominant vibration frequencies using frequency-domain analysis.
- Validate the method on more than one physical system.
- Investigate real-world issues such as interference, detector limitations, transient behavior, and optical alignment.
- Document what worked, what failed, and how the prototype could be improved toward a more rigorous photonic measurement system.

1.1 Design Philosophy

The project was developed as an engineering experiment rather than a single “working demo.” Measurements were repeatedly modified when unexpected spectral components or unstable optical behavior appeared. This iterative process — observe, hypothesize, modify, and retest — became a central part of the project and provided practical experience in the interaction between optics, electronics, embedded systems, and digital signal processing.

2. System Concept and Measurement Principle

The system uses an optical lever / reflected-intensity principle. A laser illuminates the vibrating target. Mechanical motion changes the position and/or coupling of the reflected beam at the receiver. The detector converts the changing optical power into an electrical signal that can be sampled by the embedded system.

$$\textit{Mechanical motion} \rightarrow \textit{reflected-beam variation} \rightarrow \textit{optical intensity variation} \rightarrow \textit{electrical signal} \rightarrow \textit{ADC samples} \rightarrow \textit{FFT}$$

2.1 Knife-Edge Optical Alignment

A major practical improvement was to avoid placing the laser spot fully inside the detector’s active region. When the entire spot remains on the detector, small target motion may produce very little change in total received power. Instead, the reflected spot was positioned close to the edge of the sensitive area. In this “knife-edge” arrangement, a small beam displacement produces a much larger change in the fraction of optical power falling on the detector. This increases sensitivity to motion without requiring a position-sensitive detector.

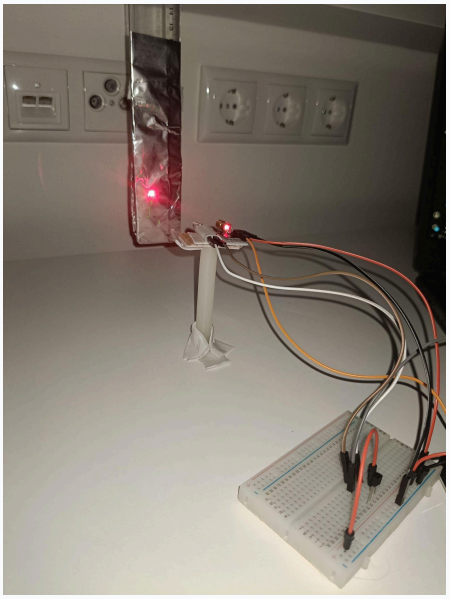
2.2 What the Prototype Measures

The prototype primarily measures frequency, not absolute displacement. The ADC amplitude depends on several uncontrolled factors, including detector responsivity, optical alignment, reflective surface, distance, ambient light, and the electrical front end. Therefore, ADC amplitude is treated as a relative optical signal. The most reliable quantity extracted in this work is the frequency of repeatable signal variation.

3. Hardware and Experimental Setup

Component	Role in the system
Laser module	Continuous optical source. The Arduino code keeps the laser control pin HIGH during acquisition.
Photodiode with NPN transistor amplification stage	Converts variations in received optical intensity into an amplified electrical signal for ADC acquisition.
Arduino-compatible microcontroller	Performs ADC sampling, sample buffering, timing control, and serial transfer.
Breadboard and jumper wiring	Rapid prototype interconnection.
Grounded aluminum shield	Experimental EMI shielding around the detector/sensitive wiring.
Metal ruler	Cantilever structure used for resonance measurements.
€2 coin	Added tip mass used to alter the mechanical response of the ruler.
DC motor	Rotating/mechanically vibrating test source.
Piezo / buzzer	Controlled excitation source; code example commands 125 Hz using tone().
Laptop with Python	Data capture, visualization, FFT processing, and automatic plot saving.

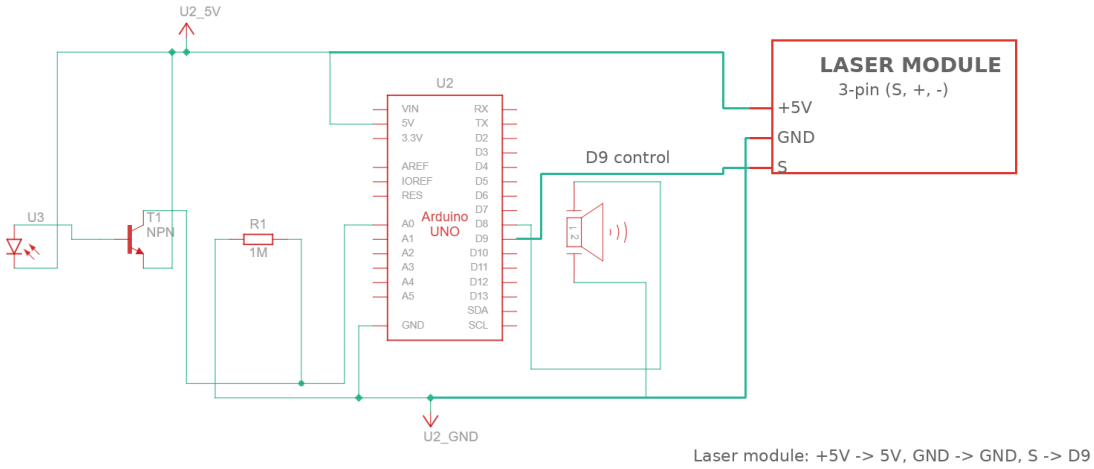
COMPLETE HARDWARE SETUP



3.1 Electrical Connections

The confirmed prototype connections used in the final acquisition setup are summarized below. The laser is a three-pin module: +5 V is connected to the 5 V rail, GND to the common ground, and the signal/control pin (S) to Arduino digital pin D9. The photodiode signal is coupled to an NPN transistor amplification stage, and the resulting analog signal is acquired through Arduino analog input A0, while the piezo/buzzer is driven from D8. The grounded aluminum shield uses the same common reference. The electrical schematic below reproduces the implemented wiring.

Connection	Arduino node	Notes
Laser control	Digital pin 9	Held HIGH for a continuous beam during acquisition.
Piezo/buzzer	Digital pin 8	Optional controlled excitation; tone() used in the provided test code.
Optical detector signal	Analog input A0	ADC input used for vibration signal acquisition.
Common reference	Arduino GND	Electrical reference; also used for the experimental grounded shield.



Final prototype electrical schematic including the three-pin laser module (S, +5 V, GND), optical detector branch, Arduino UNO, and piezo/buzzer.

3.2 Detector Dynamic Range

The detector produced a relatively small useful variation compared with the full 10-bit ADC range. In many tests, the vibration-related change was on the order of roughly 10–20 ADC counts, while the operating point could sit near the upper end of the 0–1023 range. This limited the available headroom and made the system sensitive to alignment, noise, and clipping. The project therefore relied more strongly on frequency-domain repeatability than on absolute amplitude accuracy.

4. Embedded Data Acquisition

The acquisition firmware was designed to reduce timing disturbances from serial communication. Instead of reading one sample and immediately printing it over USB serial, the microcontroller first stores a complete block of samples in RAM. Only after the acquisition is finished are the values transmitted to the computer. This separation keeps serial transmission outside the time-critical sampling loop.

$$Sampling\ interval = 1000\ \mu s \Rightarrow nominal\ sampling\ frequency\ Fs \approx 1000\ Hz$$

Each acquisition stores 550 samples. Before storage begins, 50 dummy ADC reads are performed to allow the analog input to settle. During analysis, the first 50 stored samples are also discarded, leaving 500 samples for the FFT. This provides a simple form of transient rejection while preserving a convenient analysis length.

4.1 Timing Strategy

```
unsigned long nextSampleTime = micros();

for (int i = 0; i < numSamples; i++) {
    while (micros() - nextSampleTime < sampleInterval) {}
    nextSampleTime += sampleInterval;
    samples[i] = analogRead(analogPin);
}
```

Using an accumulated target time (nextSampleTime += sampleInterval) is preferable to simply placing delay(1) after each measurement, because the schedule does not intentionally add the ADC execution time to every interval. However, the report describes the rate as nominally 1 kHz rather than perfectly calibrated 1 kHz, because micros()-based timing and analogRead() execution were not independently characterized with laboratory timing equipment.

4.2 Sampling Range and Nyquist Criterion

$$Nyquist\ frequency\ fN = Fs / 2 = 1000 / 2 = 500\ Hz$$

A 1 kHz sample rate makes 500 Hz the theoretical Nyquist boundary. This is not equivalent to saying that the system measures every vibration up to 500 Hz with 2 Hz accuracy. Practical usable bandwidth is lower because detector bandwidth, analog response, anti-alias filtering, signal-to-noise ratio, timing uncertainty, and behavior close to Nyquist all matter.

5. Signal Processing and FFT Analysis

Python is used to trigger a capture, read the 550 samples over serial, discard the initial transient region, remove the DC component, calculate the FFT, identify the strongest spectral component above 5 Hz, and save a high-resolution plot automatically. The final script also timestamps each output image to support repeated experiments.

5.1 DC Removal

```
signal = np.array(valid_data)
signal = signal - np.mean(signal)
```

Removing the mean suppresses the large DC component associated with the average received optical intensity. The analysis then focuses on changes around that operating point, which are the components associated with motion and interference.

5.2 Frequency Resolution

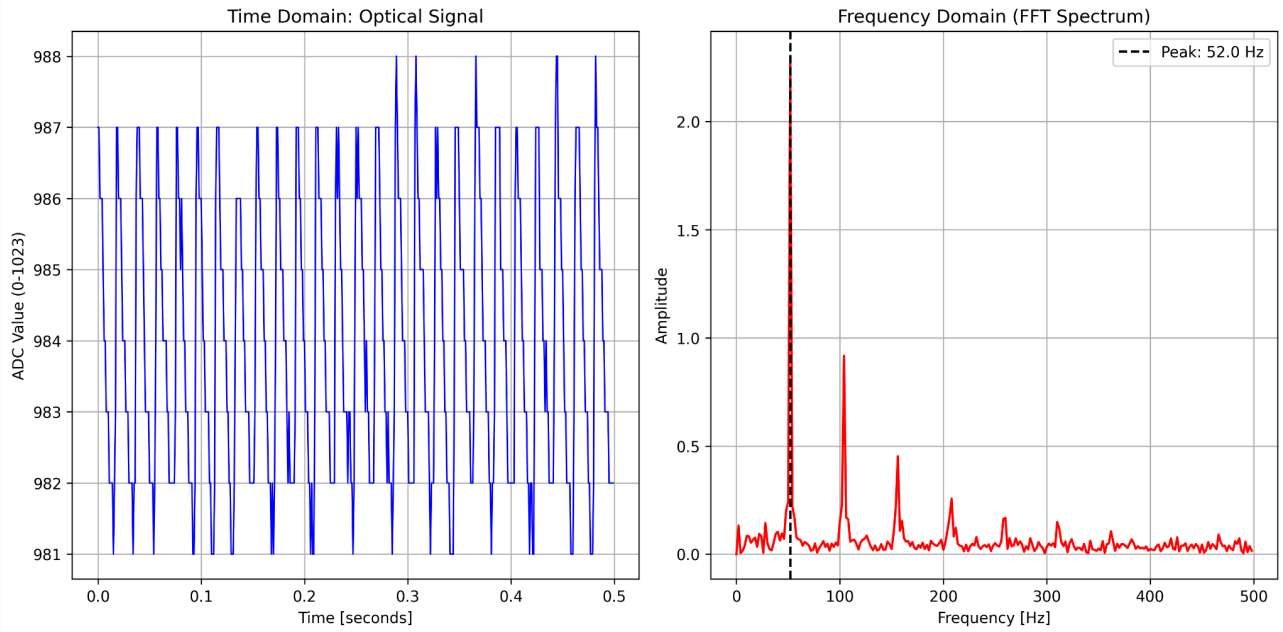
$$\Delta f = Fs / N = 1000 / 500 = 2\ Hz$$

The 2 Hz value is the FFT bin spacing (frequency resolution of the chosen analysis window), not an independently calibrated ±2 Hz absolute accuracy specification. A true vibration can lie between FFT bins, and additional uncertainty can arise from sampling-clock error, spectral leakage, noise, limited record length, and peak-selection method.

5.3 Peak Selection

The script ignores frequencies below 5 Hz when selecting the dominant peak. This reduces the probability that slow drift or residual startup behavior is incorrectly reported as the vibration frequency. The selected peak should still be interpreted alongside the time-domain waveform and repeated measurements rather than treated automatically as the true physical resonance.

EXAMPLE TIME-DOMAIN AND FFT PLOT-Motor



6. Experimental Tests and Results

Four experimental cases were used to test whether the optical measurement chain could recover meaningful frequency information under different mechanical conditions.

Test	Purpose	Measured dominant component	Interpretation
Cantilever ruler	Measure free vibration / resonance	16 Hz	Dominant mechanical vibration component of the clamped ruler.
Ruler + €2 coin	Explore added-mass effect	14 Hz	Expected trend: added effective mass lowers natural frequency; this run was less clean and should be described honestly.
DC motor	Detect periodic rotating/mechanical motion	52 Hz (reported test)	Higher-order components near 104 Hz and 156 Hz were also observed.
Piezo / buzzer	Controlled excitation reference	Commanded: 125 Hz; measured: 124 Hz	Checks whether a known/commanded

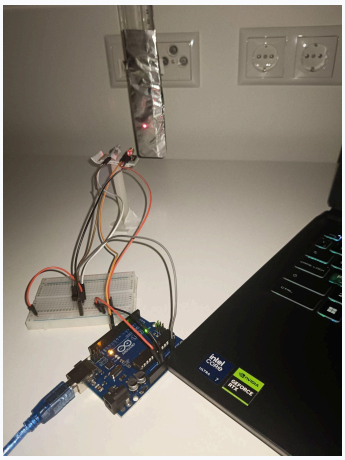
			excitation can be recovered optically.

6.1 Cantilevered Ruler

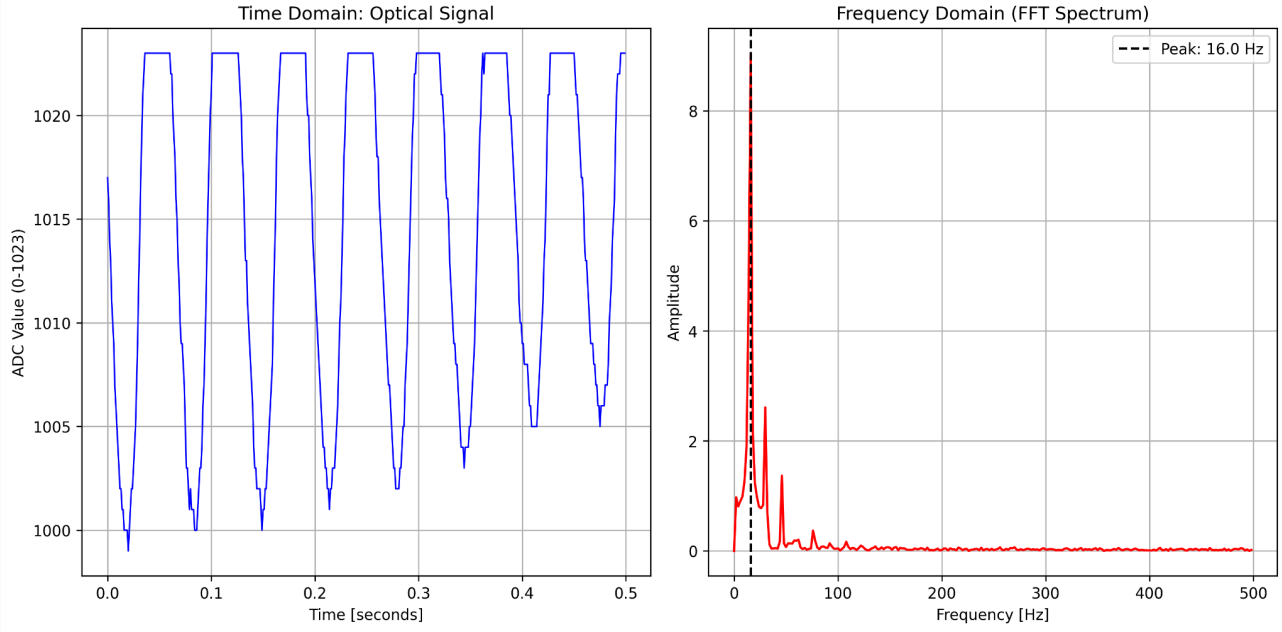
A metal ruler was clamped so that a free length could oscillate as a cantilever. The laser was directed toward the moving region and the reflected beam was aligned close to the edge of the detector. The ruler was displaced and released, producing a decaying mechanical oscillation. The optical waveform was captured and transformed using the same analysis pipeline used for all experiments.

Final reported dominant frequency: 16 Hz

RULER TEST SETUP



RULER FFT RESULT



6.2 Added-Mass Test: Ruler + €2 Coin

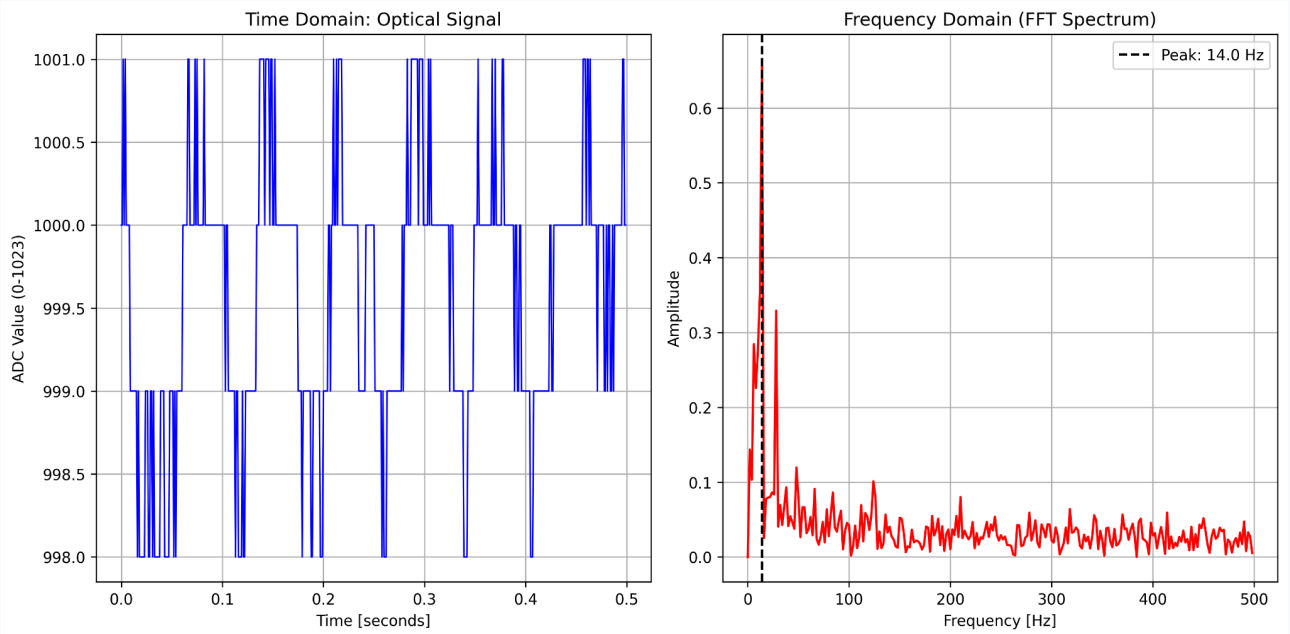
A €2 coin was attached near the free end of the ruler to change the effective mass of the vibrating structure. For a simplified mechanical oscillator, the natural frequency follows the qualitative relationship

$$fn \approx (1 / 2\pi) \cdot \sqrt{(k / meff)}$$

so an increase in effective mass is expected to reduce the resonance frequency if stiffness and boundary conditions remain approximately unchanged. The experiment showed the expected direction of change, but the result was less clean than the unloaded-ruler measurement. It is therefore treated as an exploratory structural-loading test rather than a calibrated quantitative validation.

Loaded-ruler dominant frequency: 14Hz

RULER + COIN COMPARISON



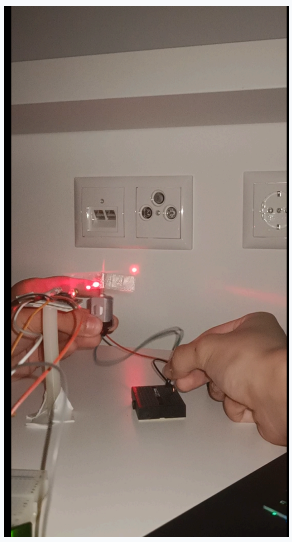
6.3 DC Motor

The optical system was also used to observe a DC motor. A dominant spectral component around 52 Hz was detected, together with higher-order components around 104 Hz and 156 Hz. These components demonstrate that the optical chain can recover periodic motion from a system very different from the cantilevered ruler.

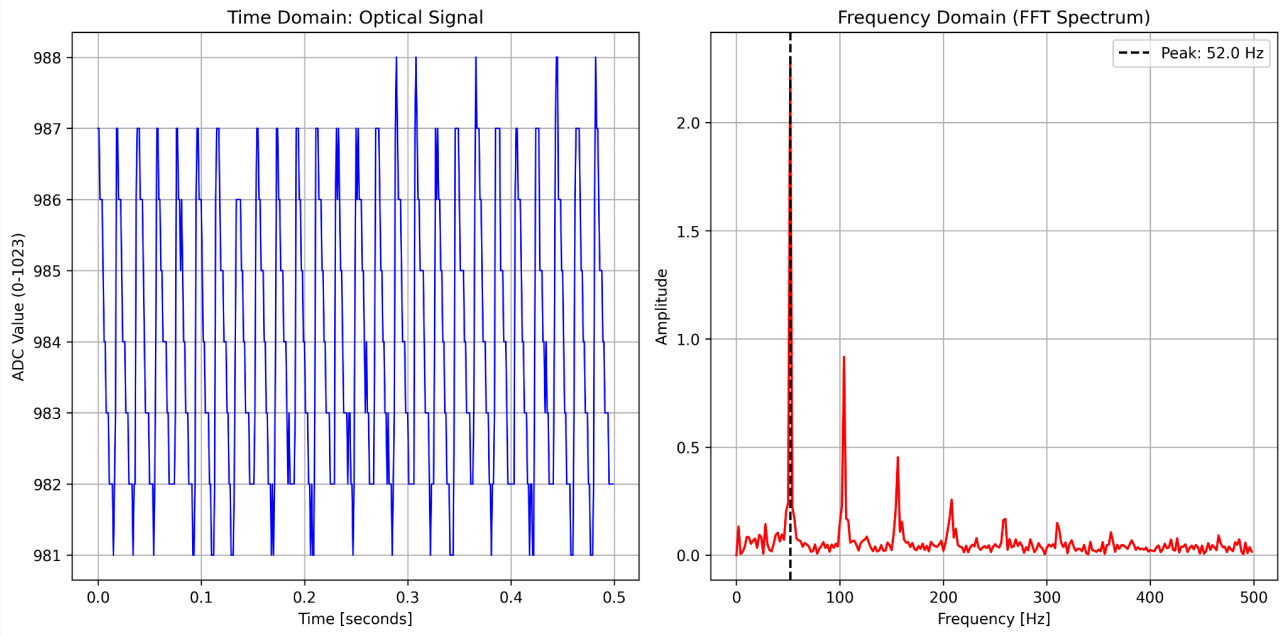
$$\textit{If one optical modulation cycle corresponds to one full revolution: } RPM \approx 52 \times 60 = 3120\ RPM$$

The RPM conversion is only valid when the optical geometry produces exactly one modulation cycle per revolution. If the target geometry creates multiple bright/dark events per revolution, the measured frequency must be divided by the number of events per revolution. For that reason, 52 Hz is the primary directly measured result; the RPM value is a conditional interpretation.

MOTOR TEST SETUP



MOTOR FFT RESULT

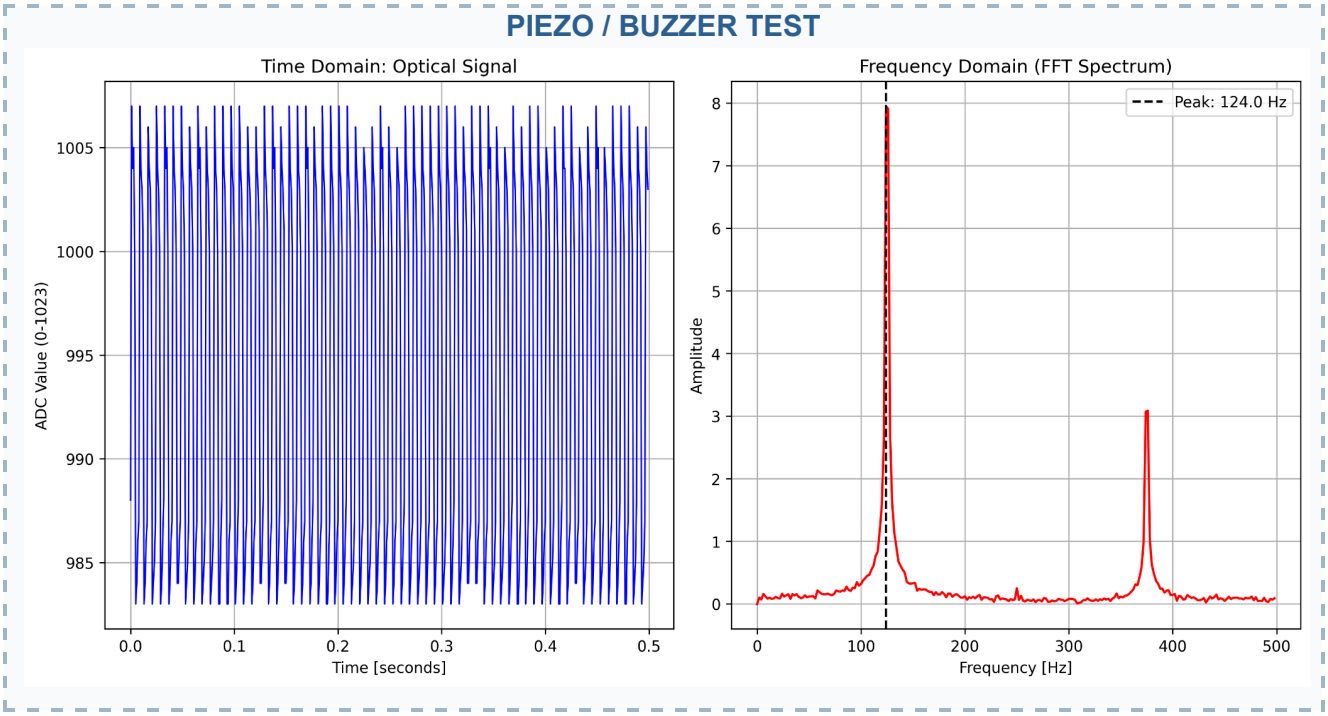


6.4 Piezo / Buzzer Excitation

A piezo/buzzer source was used as a controlled excitation case. In the supplied firmware, tone(buzzerPin, 125) commands a 125 Hz electrical drive. This experiment is useful because the excitation frequency is known by design, although the mechanical motion transferred to the measured object can still depend on mounting and coupling.

Measured optical peak during piezo/buzzer test:124 Hz

The excitation was commanded at 125 Hz, while the FFT reported the nearest dominant frequency bin at 124 Hz because the analysis used a 2 Hz spectral bin spacing.



7. Engineering Challenges and Troubleshooting

The most valuable engineering work occurred while the prototype was not behaving ideally. The following problems were investigated during development.

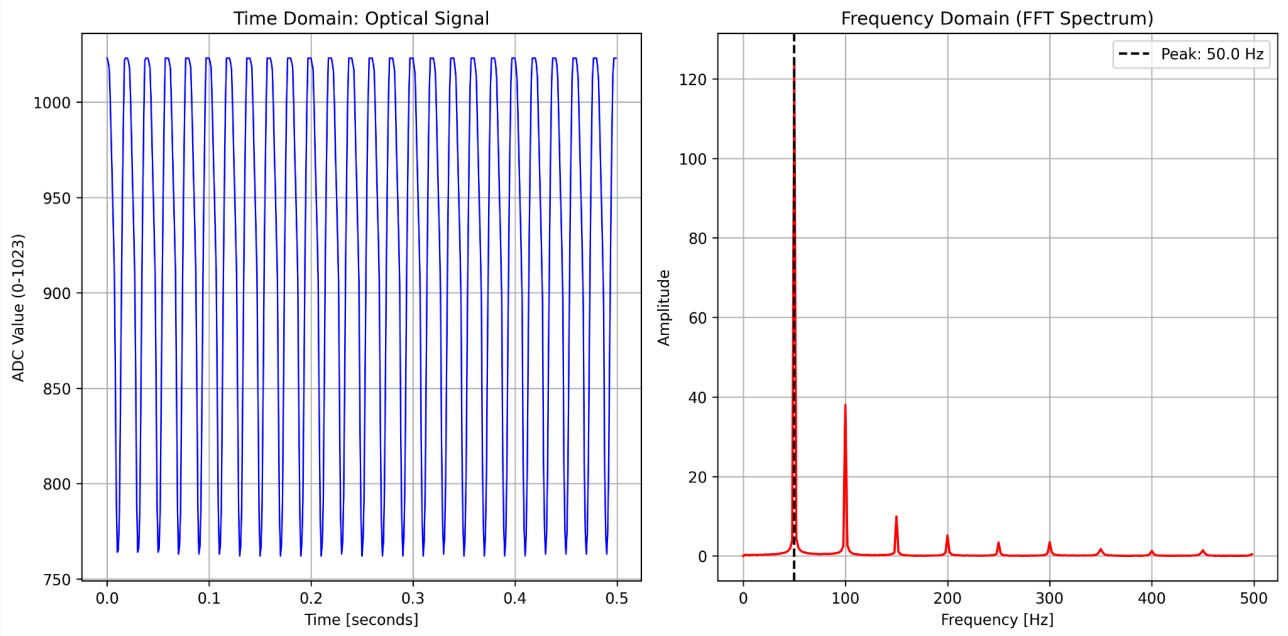
Problem	Observation	Engineering response	Outcome
Limited usable signal swing and ADC headroom	Useful vibration changes were often only ~10–20 ADC counts, and the baseline could sit close to the top of the ADC range.	Improved optical alignment using the detector edge; relied on repeated frequency-domain measurements rather than amplitude calibration.	The vibration frequency could still be extracted, but amplitude accuracy remained limited.
50 Hz mains-related component	A strong spectral component at 50 Hz repeatedly appeared in the measurement environment; additional higher-frequency components were sometimes present.	Tested grounding and conductive shielding around the receiver/sensitive wiring.	Grounded aluminum shielding reduced the observed 50 Hz component by approximately 70% in the tested configuration.
Uncertain origin of harmonics	Higher-order spectral components appeared during the 50 Hz interference tests.	The photodetector signal was amplified using an NPN transistor. Nonlinear transistor operation, clipping, lighting modulation, and electrical pickup were considered possible contributors to the observed harmonics.	Because these mechanisms were not independently isolated, the report does not assign the harmonics to a single cause.

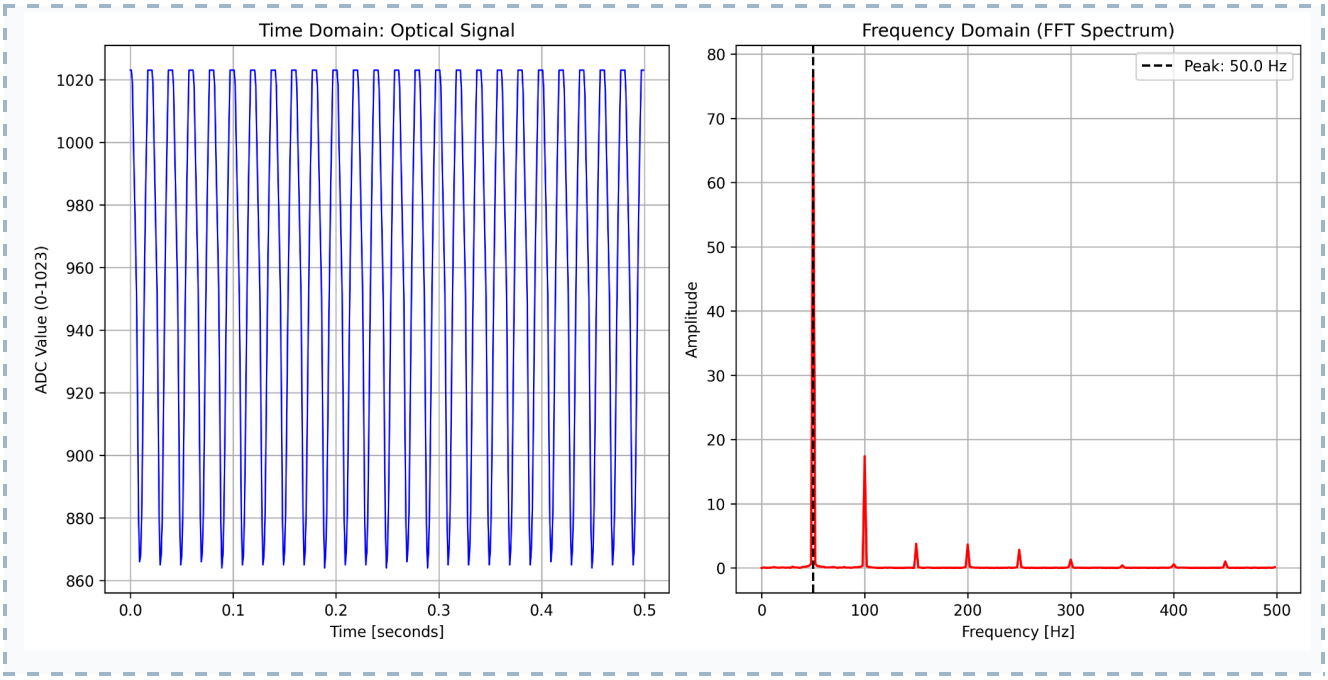
Resistor/loading experiments degraded the signal	Higher resistance values increased sensitivity to noise and parasitic effects and reduced the quality of the useful vibration signal.	Removed the problematic loading arrangement in the final working prototype and prioritized the configuration that preserved the detectable frequency content.	Working frequency extraction was restored, while the lack of a designed analog front end remained a known limitation.
Startup / low-frequency artifacts	Early samples and settling behavior could generate misleading low-frequency spectral energy.	Performed 50 dummy ADC reads before acquisition and discarded the first 50 stored samples in Python.	The analyzed record became more stable and low-frequency startup artifacts were reduced.
Optical alignment sensitivity	Small setup movements strongly affected the signal amplitude.	Used a knife-edge alignment and mechanically stabilized the laser/receiver where possible.	Improved sensitivity to vibration, but the prototype remained alignment-sensitive.
Serial timing risk	Printing each sample during acquisition would disturb sampling timing.	Buffered samples first, then transmitted them after capture.	Serial communication no longer occupied the time-critical acquisition loop.

7.1 Investigation of the 50 Hz Component

Because the European power system operates at 50 Hz, a 50 Hz component observed in a laboratory prototype is naturally suspicious. However, a photodetector can receive 50/100 Hz optical modulation from room lighting, while high-impedance wiring can also pick up electrical interference. The reduction achieved by grounded conductive shielding supports the presence of electrical coupling, but it does not prove that electrical pickup was the only source. A future controlled test should independently vary room lighting, shielding, cable geometry, and power sources to separate these mechanisms.

INTERFERENCE BEFORE / AFTER SHIELDING





7.2 Software Filtering Experiments

During troubleshooting, notch-filter approaches were explored for suppressing known mains-related components. This is useful when a deterministic interference frequency must be rejected, but filtering can also introduce transient behavior and can remove real mechanical information if a true vibration lies near a notch. For that reason, the final report separates hardware noise reduction from optional software filtering and does not use digital filtering as a substitute for understanding the measurement chain.

8. Limitations and Measurement Uncertainty

The prototype successfully recovers useful frequency information, but it is not a calibrated vibration instrument. The following limitations define the current measurement quality:

- No absolute displacement calibration: ADC counts are not converted to micrometers or angular displacement.
- Limited optical-detector dynamic range and ADC headroom; some measurements operate close to the upper rail.
- No dedicated low-noise transimpedance amplifier optimized for the photodiode.
- No analog anti-aliasing filter before the ADC.
- Nominal sampling timing is generated using micros() rather than a hardware timer validated against external equipment.
- Frequency estimate is quantized by the finite FFT record; the 2 Hz bin spacing is not identical to absolute measurement accuracy.
- Optical geometry is sensitive to alignment, distance, reflectivity, and ambient illumination.
- The motor RPM conversion requires knowledge of the number of optical modulation events per revolution.
- The 50 Hz interference mechanism was reduced but not fully isolated into optical versus electrical contributions.

9. Future Improvements

The project provides a clear roadmap for upgrading the prototype from a frequency-demonstration system toward a more rigorous optical instrument.

Dedicated transimpedance amplifier (TIA): Convert photodiode current to voltage with controlled gain and bandwidth. This should increase usable signal amplitude, improve headroom, and make the detector behavior predictable.

Mid-supply bias / analog headroom: Bias the front end so the signal is centered within the ADC range rather than near 1023, reducing clipping and increasing effective dynamic range.

Differential / split photodetection: Use two photodiodes or a quadrant/position-sensitive detector so beam movement is measured differentially. Common-mode optical intensity changes can then be reduced.

Analog anti-aliasing filter: Low-pass the signal before sampling to suppress components above the usable bandwidth that could alias into the FFT.

Hardware-timer sampling: Use a timer interrupt or ADC auto-trigger to create a more accurately defined sampling frequency with lower jitter.

Higher-resolution ADC: A 12- or 16-bit converter could better resolve the small optical amplitude changes when the analog front end is designed appropriately.

Longer acquisition window: Increasing N improves FFT bin spacing (for the same Fs), allowing closer spectral components to be separated, at the cost of memory and acquisition duration.

Windowing and peak interpolation: A Hann window and sub-bin peak interpolation can improve spectral interpretation when the vibration frequency does not fall exactly on an FFT bin.

Controlled interference testing: Perform tests with room lights on/off, battery power, different shield geometries, and short/shielded cables to isolate the 50 Hz coupling path.

Absolute calibration: Use a reference displacement source or calibrated accelerometer/vibrometer to map optical signal amplitude to physical motion and quantify error.

Automated experimental logging: Save raw samples, metadata, configuration, and derived peaks into CSV/JSON for traceable, repeatable characterization.

10. Relevance to Photonic Sensing and Structural Monitoring

This prototype is intentionally much simpler than professional fiber-optic sensing systems, but it demonstrates a related measurement workflow: a mechanical quantity modifies an optical signal, the optical response is converted into electrical data, and frequency-domain analysis extracts information about the physical system. This same chain — optical sensing, acquisition, evaluation, and interpretation — is central to many photonic sensing applications.

A particularly relevant research example is Fraunhofer HHI's FiberLab, which develops fiber-optic sensors, sensor networks, and systems for the acquisition and evaluation of sensor data. HHI's CASSIOPEIA project applies fiber-optic sensing to Structural Health Monitoring of buildings in earthquake regions and explicitly uses structural vibrations and eigenfrequencies as measurement parameters. The present project does not reproduce that technology: it uses free-space reflected light and a simple photodetector instead of fiber Bragg gratings and a multiplexed professional interrogation system. The connection is therefore conceptual and methodological rather than a claim of equivalent performance.

11. Conclusion

A low-cost non-contact optical vibration sensing prototype was designed, built, debugged, and tested on multiple mechanical systems. The setup uses reflected laser light to convert motion into intensity variation, a photodetector and Arduino-compatible acquisition stage to digitize the signal, and Python/FFT

processing to extract dominant frequency components. The final workflow uses nominal 1 kHz sampling, block buffering, transient rejection, DC removal, and automated plotting. With 500 analyzed samples, the FFT bin spacing is 2 Hz.

The system successfully identified the dominant vibration behavior of a cantilevered ruler and a DC motor, with the motor experiment showing a strong component near 52 Hz and higher-order spectral components. Additional experiments explored added mechanical mass and controlled piezo/buzzer excitation. The project also demonstrated how strongly practical optical measurements can be affected by detector operating point, parasitic loading, environmental interference, alignment, and startup behavior. Grounded shielding reduced the observed 50 Hz mains-related component by roughly 70% in the tested setup.

The most important result is the complete engineering workflow developed around the measurement: build, observe, analyze, isolate problems, modify the hardware/software, repeat the experiment, and state the remaining limitations. A future version using a designed photodiode transimpedance front end, differential detection, controlled analog filtering, and hardware-timed acquisition could provide substantially better sensitivity and quantitative accuracy.

Technical Skills Demonstrated

Optical sensing • Photodetection • Laser alignment • Analog measurement • Embedded C/C++ • ADC acquisition • Timing and buffering • Python • NumPy/SciPy • FFT and spectral analysis • EMI troubleshooting • Grounding and shielding • Experimental characterization

Appendix A — Arduino Acquisition Code

The following is the acquisition code provided for the final test workflow. It keeps the laser continuously on, optionally drives a buzzer at 125 Hz, performs settling reads, captures 550 ADC samples at a nominal 1 ms interval, buffers them, and transmits them only after acquisition.

```
const int laserPin = 9;
const int buzzerPin = 8;
const int analogPin = A0;
const int numSamples = 550;

// 1000 microseconds = nominal 1000 Hz sampling rate
const unsigned long sampleInterval = 1000;
int samples[numSamples];

void setup() {
  Serial.begin(115200);
  pinMode(laserPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  // Continuous optical beam during acquisition
  digitalWrite(laserPin, HIGH);

// Uncomment the following line only for the controlled 125 Hz
// piezo/buzzer validation experiment.

// Enable only during the piezo/buzzer validation experiment
// tone(buzzerPin, 125);

}

void loop() {
  if (Serial.available() > 0) {
    char cmd = Serial.read();
    if (cmd == 's' || cmd == 'S') {
      // Dummy ADC reads / settling period
      for (int i = 0; i < 50; i++) {
        analogRead(analogPin);
        delay(2);
```

```
      }

      captureData();
      dumpData();
    }
  }
}

void captureData() {
  unsigned long nextSampleTime = micros();

  for (int i = 0; i < numSamples; i++) {
    while (micros() - nextSampleTime < sampleInterval) {}
    nextSampleTime += sampleInterval;
    samples[i] = analogRead(analogPin);
  }
}

void dumpData() {
  for (int i = 0; i < numSamples; i++) {
    Serial.println(samples[i]);
  }
}
```

Appendix B — Python Capture and FFT Analysis Code

The script below triggers capture over serial, receives 550 samples, discards the first 50 stored samples, removes the DC offset, calculates the FFT, detects the strongest component above 5 Hz, and automatically saves the plot.

```
import time
import serial
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq

SERIAL_PORT = 'COM9'
BAUD_RATE = 115200
TOTAL_SAMPLES = 550
FS = 1000.0

def capture_and_analyze():
    print(f"Connecting to {SERIAL_PORT}...")
    try:
        ser = serial.Serial(SERIAL_PORT, BAUD_RATE, timeout=2)
        time.sleep(2)
        ser.reset_input_buffer()
        print("Triggering data capture...")
        ser.write(b's')

        raw_data = []
        for _ in range(TOTAL_SAMPLES):
            line = ser.readline().decode('utf-8').strip()
            if line.isdigit():
                raw_data.append(int(line))

        ser.close()

        if len(raw_data) != TOTAL_SAMPLES:
            print(f"Error: Received {len(raw_data)} samples instead of {TOTAL_SAMPLES}.")
            return

        # Discard first 50 stored samples
        valid_data = raw_data[50:]
        analyzed_samples = len(valid_data)

        signal = np.array(valid_data)
        signal = signal - np.mean(signal)
```

```
        t = np.arange(analyzed_samples) / FS

        yf = fft(signal)
        xf = fftfreq(analyzed_samples, 1 / FS)[:analyzed_samples // 2]
        amplitude = (2.0 / analyzed_samples) * np.abs(yf[0:analyzed_samples // 2])

        valid_indices = np.where(xf > 5)[0]
        peak_idx = valid_indices[np.argmax(amplitude[valid_indices])]
        dominant_freq = xf[peak_idx]

        plt.figure(figsize=(12, 6))

        plt.subplot(1, 2, 1)
        plt.plot(t, valid_data, linewidth=1)
        plt.title('Time Domain: Optical Signal')
        plt.xlabel('Time [seconds]')
        plt.ylabel('ADC Value (0-1023)')
        plt.grid(True)

        plt.subplot(1, 2, 2)
        plt.plot(xf, amplitude, linewidth=1.5)
        plt.title('Frequency Domain (FFT Spectrum)')
        plt.xlabel('Frequency [Hz]')
        plt.ylabel('Amplitude')
        plt.axvline(x=dominant_freq, linestyle='--', label=f'Peak: {dominant_freq:.1f} Hz')
        plt.legend()
        plt.grid(True)

        plt.tight_layout()
        filename = f"fft_result_{time.strftime('%H%M%S')}.png"
        plt.savefig(filename, dpi=300, bbox_inches='tight')
        print(f"Graph saved automatically as: {filename}")
        print("Done")
        plt.show(block=False)
        plt.pause(10)
        plt.close()

    except Exception as e:
        print(f"Connection failed: {e}")

if __name__ == '__main__':
    capture and analyze()
```

References

[1] Fraunhofer Heinrich Hertz Institute (HHI), “FiberLab – Fiber Optic Sensing with Femtosecond Lasers.” https://www.hhi.fraunhofer.de/en/departments/fs/research-groups/fiberlab.html

[2] Fraunhofer Heinrich Hertz Institute (HHI), “CASSIOPEIA – Centralized Fiber-Optic Sensor System for Building Monitoring in Earthquake Regions.” https://www.hhi.fraunhofer.de/en/departments/fs/projects/cassiopeia.html

[3] NumPy/SciPy documentation for discrete Fourier transform / FFT processing and scipy.fft.

[4] Arduino documentation .